



Racial/Ethnic and Gender Differences in Associations of Medication-Assisted Therapy and Reduced Opioid Use between Outpatient Treatment Admission and Discharge

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ABSTRACT

Medication-assisted therapy (MAT) for opioid use disorders is an effective treatment strategy. Racial/ethnic and gender disparities in MAT utilization have been documented, but less is known about disparities in MAT outcomes. We used the Treatment Episodes Dataset–Discharges (TEDS-D; 2015–2017) to identify outpatient treatment episodes with heroin or illicit opioids indicated at admission ($n = 232,547$). We used multivariate logistic regression to model the association between MAT and a reduction in opioid use between treatment admission and discharge. We explored moderation by race/ethnicity and gender by including an interaction term. We identified a strong moderating effect of race/ethnicity and gender. American Indian/Alaska Native (AI/AN) women demonstrated the strongest association between MAT (versus no MAT) and a reduction in opioid use ($aOR = 6.05$, $95\% CI = 4.81–7.61$), while White men demonstrated the weakest association ($aOR = 2.78$, $CI = 2.70–2.87$). Our findings could inform changes in clinical MAT settings that are based on harm reduction and the incremental transition from illicit opioids to medication-assistance among a diverse opioid use disorder population.

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Introduction

Rates of heroin and opioid dependence and abuse have reached record highs over the past decade (Jones et al. 2015; Lipari and Hughes 2015), resulting in an epidemic of opioid-related overdose deaths (Calcaterra, Glanz, and Binswanger 2013; Rudd et al. 2016). While the rates of heroin and opioid use have increased across most racial/ethnic and gender groups, the increase has been particularly pronounced among Whites (Cicero, Ellis, and Surratt 2014; Friedman et al. 2019; Martins et al. 2015; Phillips, Ford, and Bonnie 2017) and women (Back et al. 2011; Campbell et al. 2010; Serdarevic, Striley, and Cottler 2018). Rates of opioid-related overdose deaths have also increased overall (Seth et al. 2018) and are expected to continue increasing by nearly 150% by the year 2025 (Chen et al. 2019). Parallel to the increase in opioid use, opioid-related overdose deaths have also risen most drastically among Whites (Rudd et al. 2014; Unick et al. 2013) and women (Evans et al. 2015; Mack, Jones, and Paulozzi 2013). As a result, overdose deaths have contributed substantially to the recent increase in all-cause mortality rates among Whites (Roux Diez 2017;

Stein et al. 2017). Addressing the opioid crisis, Dasgupta, Beletsky, and Cicarone (2017) recently recognized the real and troubling trends among Whites, while also making a compelling case for continued investigations into racial/ethnic opioid abuse and treatment disparities.

Medication-assisted therapy (MAT) (methadone or buprenorphine) for opioid use disorders (OUD) is effective at reducing withdrawal symptoms and managing opioid abstinence (Connery 2015). Buprenorphine and methadone differ in terms of federal prescribing regulations (US Department of Health and Human Services 2019), yet both substantially reduce persistent cravings and withdrawal symptoms with similar efficacy (Mattick et al. 2004). Additionally, MAT has been shown to decrease overdose deaths and rates of infectious diseases associated with OUD, reduce criminal activity, and strengthen social functioning (Volkow et al. 2014). Despite the benefits of MAT, only 19% to 29% of individuals with an OUD receive MAT or other opioid-specific treatment (Krawczyk et al. 2017; L; Wu, Zhu, and Swartz 2016).

MAT utilization differs by race/ethnicity and gender, but findings are nuanced and the direction of effects vary

based on multiple study-related factors, including the specific opioid under consideration, the size and provincial nature of the sample, and treatment modality. For example, Hansen et al. (2016) found that buprenorphine treatment in New York City increased at a significantly higher rate in areas with the highest incomes and the lowest percentage of Blacks and Hispanic between 2004 and 2013. Lundgren et al. (2001) demonstrated lower likelihoods of methadone-specific services uptake among Black and Hispanic injection drug users (IDU) in Massachusetts, compared to White IDUs. In contrast, a recent report using the national 2014 Treatment Episodes Dataset-Admissions (TEDS-A) found that White heroin users were the least likely to receive MAT, compared to their Black and Hispanic counterparts (Krawczyk et al. 2017). While the Krawczyk report did not include American Indian/Alaska Native (AI/AN) MAT participants, Duvivier et al. (2017) found that substantial investments in substance abuse services targeting AI/AN through the Indian Health Service (IHS) have resulted in expanded access to MAT and other harm reduction strategies. Finally, although OUDs tend to be more prevalent among women than men, the former are less likely to utilize treatment services (Back et al. 2010).

Evidence is mixed regarding disparities in OUD treatment outcomes, and study findings are dependent upon the type of opioid, route of administration (ingested, smoked, snorted, or injected), and treatment regimens under investigation. Potter et al. (2013) found that Whites were more likely than African Americans to abstain entirely from opioids at the conclusion of methadone or buprenorphine treatment. Several other studies have also demonstrated a lower likelihood of positive OUD treatment outcomes among African Americans compared to Whites (Banta-Green et al. 2009; Mennis and Stahler 2016; Stahler and Mennis 2018; Weinstein et al. 2017; LT Wu et al. 2010). Banta-Green et al. (2009) found no significant difference in OUD treatment outcomes between AI/AN and Whites, but few other studies have reported on AI/AN OUD treatment outcomes and racial/ethnic group differences. Gender differences are not as apparent in the OUD treatment outcomes literature, as several studies have found no significant gender differences in OUD treatment outcomes (Bawor et al. 2015; McHugh et al. 2013; Potter et al. 2013).

Definitions of treatment success vary and are often presented in terms of treatment completion or continued substance use following treatment. However, these outcomes fail to capture indications of a reduction in opioid use and, therefore, decreased likelihood of adverse opioid-related events (Zhang, Friedmann, and

Gerstein 2003). Fewer episodes of any opioid use are associated with a decreased risk of overdose (Brugal et al. 2002), decreased risk of accidental or unknowing injection of powerful synthetic opioids like fentanyl (Macmadu et al. 2017), fewer opportunities for the transmission of communicable diseases related to needle sharing (Bruneau et al. 2012), and a reduced risk of arrest and incarceration (Young and Havens 2012). Therefore, a reduction in the use of heroin or prescription opioids is beneficial to the overall health and safety of the user, but may be missed in conventional outcome metrics like treatment completion. Tolerating some continued opioid use or relapse while continuing participation in treatment services is an effective harm reduction strategy, primarily aimed at improving overall health, risk avoidance, and eventual transition into abstinence-based services (Veilleux et al. 2010). Objective analyses of risks and benefits may better inform OUD treatment providers as they weigh harm reduction approaches.

We sought to measure the association between MAT and a decrease in opioid use between outpatient treatment admission and discharge. We explored moderation by race/ethnicity and gender groups. We hypothesized that treatment participants who received MAT would be more likely to reduce opioid use between admission and discharge, compared to those who did not receive MAT. Importantly, the strength of the association between MAT and a reduction in opioid use would vary within racial/ethnic/gender groups, with the strongest positive association among White men and White women.

Methods

We used the Treatment Episode Dataset-Discharges dataset (TEDS-D) (Substance Abuse and Mental Health Data Archive 2018) to identify treatment episodes that had heroin or other illicit opioids indicated as a substance of abuse at treatment admission between 2015 and 2017 ($n = 232,547$). We included only the three most recent years of data because starting in 2015, TEDS-D began reporting the frequency of substance use at both treatment admission and treatment discharge, allowing for new analyses of changes between admission and discharge time points.

We did not include inpatient detoxification or inpatient residential/rehabilitation treatment settings in our sample. The use of illicit opioids is prohibited in inpatient settings and access to substances is highly controlled, although not impossible to achieve. Including inpatient treatment episodes may have increased our risk of selection bias. Thus, our sample includes only outpatient treatment episodes, as these are the cases with the most access

to opioids and the greatest opportunity to continue illicit use.

Data collection methods and TEDS data have been described elsewhere (Substance Abuse and Mental Health Services Administration 2016). In brief, TEDS is comprised of annual, repeated, cross-sectional, administrative data from treatment facilities that receive any federal funding. In addition, it is possible that treatment clients initiate treatment more than once in a year, meaning individual clients could have multiple treatment episodes recorded in any given year.

Variables

We constructed an 8-level race/ethnicity/gender variable from the TEDS race/ethnicity and gender variables. Our analytic variable included White men, White women, Black men, Black women, Latino men, Latina women, AI/AN men, and AI/AN women. We did not include several racial/ethnic groups due to either small sample sizes that resulted in model convergence problems and inestimable odds ratios (Asian/Pacific Islander, < 0.1%); or our inability to meaningfully interpret the TEDS-D group classification (other single race, 8.6%; two or more races, 2.8%).

We created a dichotomous dependent variable of a reduction in the frequency of opioid use between treatment admission and discharge. Frequency of use is defined in TEDS-D as daily use in the past month, some use in the past month, or no use in the past month. We defined a reduction in the frequency of opioid use as 1) daily use at admission and some use at discharge, 2) daily use at admission and no use at discharge, or 3) some use at admission and no use at discharge. All other combinations of frequency of use between admission and discharge were defined as an increase in the frequency of opioid use or no change in frequency. We grouped an increase in opioid use together with no change in frequency because of the clinical meaningfulness of this metric. While regular or continued opioid use may be tolerated in some MAT programs, the ultimate clinical goal is to reduce the frequency of use and eventually achieve abstinence through replacing illicit opioids with medication assistance. Therefore, patients with the same level of use at both admission and treatment would likely be considered unsuccessful or non-adherent.

Our focal independent variable was a binary indicator of whether a treatment participant received MAT during treatment. In TEDS-D, this variable is labeled “medication-assisted opioid therapy”, and indicates “if the use of opioid medications such as methadone or buprenorphine will be part of the client’s treatment”. We also considered several covariates in our analyses, some of which were

pre-categorized by TEDS, including age (18–29 [ref], 30–39, 40–49, or more than 50 years), educational attainment (less than high school [ref], high school completion, some college, or college completion), whether a treatment participant injects opioids (non-injector [ref] or injector), co-occurring abuse of alcohol (no [ref] or yes), and the year of treatment (2015 [ref], 2016, or 2017). Our outpatient treatment service type variable combined three TEDS services, including ambulatory non-intensive outpatient [ref] (treatment services including individual, family, and/or group services), ambulatory intensive outpatient (treatment services lasting two or more hours per day for three or more days per week), and ambulatory outpatient detoxification (treatment services providing for safe withdrawal). We constructed an indicator of treatment duration by combining several levels of the length of treatment stay (1–90 days [ref], 91–180 days, 181–365 days, or more than 365 days). Our opioid type variable included prescription/non-heroin opioids only [ref] (prescription opioids, non-prescription use of methadone, or other opiates with morphine-like effects), heroin only, or a combined variable level of both prescription/non-heroin and heroin. Our analyses used a publicly available de-identified dataset, and were exempt from review by the Northern Arizona University Institutional Review Board.

Analysis

We used SAS (v9.4) for all analyses (SAS Institute Inc., Cary, NC). We used univariate logistic regression to assess associations between a reduction in opioid use between admission and discharge time points and each covariate. In SAS, we used a procedure for generalized linear models (PROC GLIMMIX with the LSMEANS statement and SLICEDIFF option) and included an interaction term between race/ethnicity/gender and MAT to test multiple pairwise differences, generating comparisons among levels of one categorical factor (MAT versus no MAT) at a fixed level of another categorical factor (each level of race/ethnicity/gender). We presented odds ratios, 95% confidence intervals, and p-values for the association between MAT and a decrease in opioid use as a main effect, as well as within each of our eight race/ethnicity/gender groups. Our model was adjusted for age, education, outpatient service type, treatment duration, opioid type, injection status, and co-occurring alcohol abuse.

Results

The majority of outpatient treatment episodes with heroin or illicit opioids reported at admission reduced the frequency of opioid use between admission and

discharge (59%) (Table 1). Less than half of the sample (42%) received MAT. Those who received MAT had more than four times higher odds of decreasing opioid use, compared to those who did not receive MAT (crude odds ratio [cOR] = 4.20, 95% confidence interval [CI] = 4.12– 4.28).

There were clear group differences in whether clients reduced the frequency of opioid use between treatment admission and discharge between 2015 and 2017 (Figure 1). All groups that received MAT had higher proportions of reduced opioid use, compared to their

group counterparts that did not receive MAT. The lowest group proportions of reduced opioid use were identified among AI/AN men and AI/AN women who did not receive MAT in all three years (30%, AI/AN men in 2017; 34%, AI/AN women in 2017). Conversely, the highest group proportion of opioid use reduction was among Latina women in 2015 (88%). There was a modest but notable curvature downward among most groups across three years of data.

In our fully adjusted model, MAT as a main effect was associated with significantly higher odds of reducing

Table 1. Characteristics of treatment episodes with heroin or other opioids indicated as a substance of abuse at outpatient treatment admission; Univariate logistic regression modeling a decrease in the frequency of opioid use between admission and discharge (TEDS-D, 2015–2017, n = 232,547).

Variables		Opioid use frequency between treatment admission and discharge				p
		Total n = 232,547	Frequency decreased	Frequency stayed the same or increased	cOR (95% CI)	
			n = 136,459 (59%)	n = 96,088 (41%)		
n (%)		col%	col%			
Received MAT						
No	133,941 (57.60)	44.08	76.80	Ref.		
Yes	98,606 (42.40)	55.92	23.20	4.20 (4.12, 4.28)	<0.0001	
Race/ethnicity/gender						
White men	101,252 (43.54)	42.38	45.18	Ref.		
White women	74,645 (32.10)	30.25	34.73	0.93 (0.91, 0.95)	<0.0001	
AI/AN men	1,796 (0.77)	0.63	0.98	0.68 (0.62, 0.75)	<0.0001	
AI/AN women	1,921 (0.83)	0.80	0.87	0.98 (0.89, 1.07)	0.64	
Black men	15,342 (6.60)	7.01	6.01	1.24 (1.20, 1.29)	<0.0001	
Black women	8,317 (3.58)	3.87	3.16	1.31 (1.25, 1.37)	<0.0001	
Latino men	21,019 (9.04)	10.73	6.63	1.73 (1.67, 1.78)	<0.0001	
Latina women	8,255 (3.55)	4.33	2.44	1.90 (1.80, 1.99)	<0.0001	
Age group						
18–29	92,710 (39.87)	37.85	42.73	Ref.		
30–39	75,877 (32.63)	31.99	33.54	1.07 (1.05, 1.10)	<0.0001	
40–49	36,145 (15.54)	16.09	14.76	1.23 (1.20, 1.26)	<0.0001	
>50	27,815 (11.96)	14.07	8.97	1.77 (1.72, 1.82)	<0.0001	
Education						
Less than high school	59,250 (25.48)	25.21	25.85	Ref.		
High school completion	110,602 (47.56)	47.17	48.12	1.00 (0.98, 1.03)	0.62	
Some college	50,609 (21.76)	21.84	21.65	1.04 (1.01, 1.06)	0.01	
College completion	12,086 (5.20)	5.77	4.38	1.35 (1.29, 1.41)	<0.0001	
Outpatient treatment service type						
Non-intensive outpatient	176,596 (75.94)	79.02	71.57	Ref.		
Intensive outpatient	50,868 (21.87)	17.62	27.91	0.57 (0.56, 0.58)	<0.0001	
Detoxification in ambulatory setting	5,083 (2.19)	3.36	0.52	5.87 (5.35, 6.44)	<0.0001	
Treatment episode duration						
1– 90 days	122,851 (52.83)	44.17	65.13	Ref.		
91– 180 days	41,669 (17.92)	18.67	16.85	1.63 (1.59, 1.67)	<0.0001	
181– 365 days	33,915 (14.58)	16.62	11.69	2.10 (2.04, 2.15)	<0.0001	
> 365 days	34,112 (14.67)	20.54	6.33	4.78 (4.64, 4.93)	<0.0001	
Reported opioid type						
Prescription/non-heroin only	60,553 (26.04)	23.71	29.34	Ref.		
Heroin only	136,026 (58.49)	58.66	58.26	1.25 (1.22, 1.27)	<0.0001	
Both	35,968 (15.47)	17.62	12.40	1.76 (1.71, 1.81)	<0.0001	
Injection status						
Non-injector	114,802 (49.37)	48.83	50.13	Ref.		
Injector	117,745 (50.63)	51.17	49.87	1.05 (1.03, 1.07)	<0.0001	
Co-occurring alcohol abuse						
No	160,195 (68.89)	81.87	50.45	Ref.		
Yes	72,352 (31.11)	18.13	49.55	0.22 (0.21, 0.23)	<0.0001	
Year						
2015	71,943 (30.94)	32.37	28.90	Ref.		
2016	74,313 (31.96)	33.37	29.95	0.99 (0.97, 1.01)	0.61	
2017	86,291 (37.11)	34.26	41.15	0.74 (0.72, 0.76)	<0.0001	

1) MAT = Medication-assisted therapy; AI/AN = American Indian/Alaska Native; cOR = crude odds ratio 2) Prescription/non-heroin includes prescription opioids, non-prescription methadone, or any other opioids with morphine-like effects

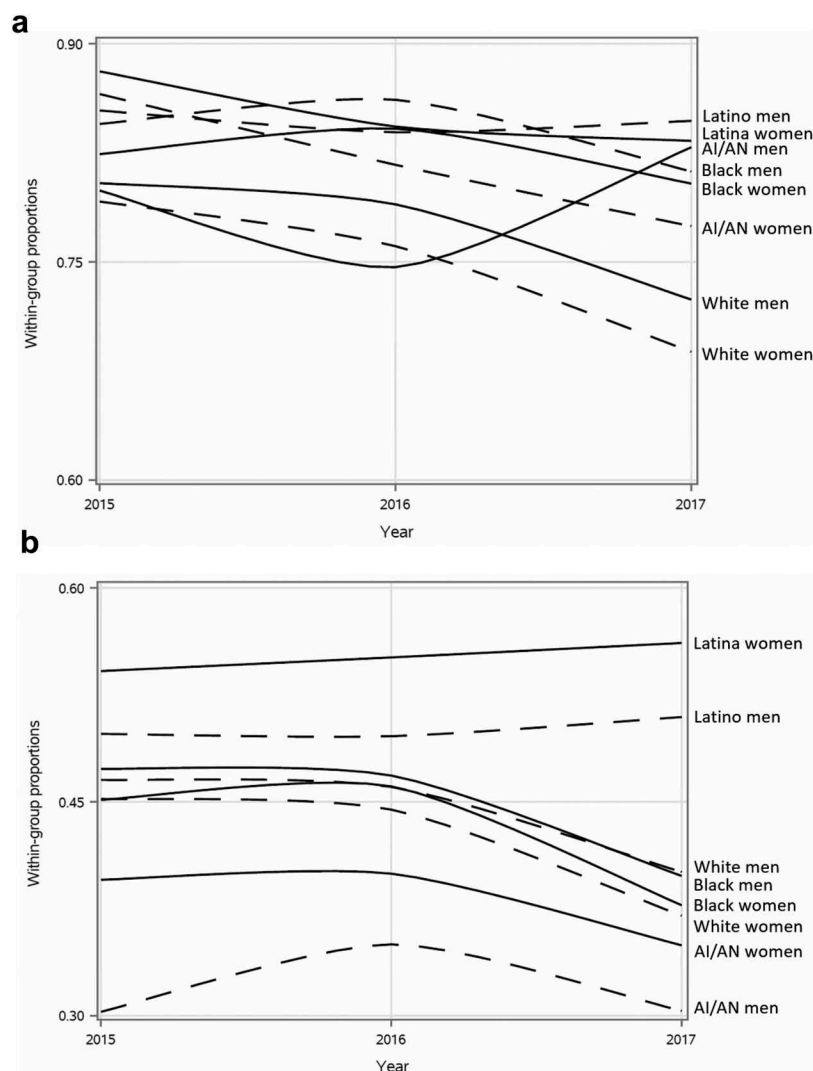


Figure 1. (A) – Within-group proportions of whether opioid use frequency decreased between treatment admission and discharge among treatment episodes that received MAT (TEDS-D, 2015–2017, $n = 98,606$). (B) – Within-group proportions of whether opioid use frequency decreased between treatment admission and discharge among treatment episodes that did not receive MAT (TEDS-D, 2015–2017, $n = 133,941$).

opioid use between treatment admission and discharge (aOR = 3.79, CI = 3.61– 3.98, $p < 0.0001$) (Table 2). Our results also demonstrated clear within-group differences in the magnitude of associations, indicating moderation by race/ethnicity/gender. Compared to those who did not receive MAT, the odds of reducing opioid use among those who received MAT were more than six times higher among AI/AN women (aOR = 6.05, CI = 4.81– 7.61), and more than five times higher among AI/AN men (aOR = 5.91, CI = 4.62– 7.54, $p < 0.0001$). The strength of the association between MAT and a reduction in opioid use was positive but weakest among White women and White men (aOR = 2.82, CI = 2.73– 2.93, $p < 0.0001$; aOR = 2.78, CI = 2.70– 2.87, $p < 0.0001$, respectively).

Discussion

We explored moderation by race/ethnicity and gender groups in the association between MAT and a reduction in opioid use among outpatient treatment participants between admission and discharge. All groups who received MAT demonstrated markedly higher proportions of a reduction in opioid use, compared to their counterparts who did not receive MAT. Overall, the group proportions of opioid use reduction were constant or decreasing over three years, with the only notable improvement observed among AI/AN men between 2016 and 2017. This general decrease in the trend of reducing opioid use – even among MAT participants – may signify a modest, short-term decrease in the efficacy of MAT services.

Table 2. Multivariable logistic regression modeling a decrease in the frequency of opioid use between treatment admission and discharge (TEDS-D, 2015–2017, $n = 232,547$).

Variables	aOR (95% CI)	p
Main effect		
MAT receipt		
No	Ref.	
Yes	3.79 (3.61, 3.98)	<0.0001
Race/ethnicity/gender * MAT interaction		
Within-group comparisons of MAT receipt (Ref = no)		
AI/AN women	6.05 (4.81, 7.61)	<0.0001
AI/AN men	5.91 (4.62, 7.54)	<0.0001
Black women	4.04 (3.63, 4.50)	<0.0001
Black men	3.97 (3.66, 4.31)	<0.0001
Latina women	2.87 (2.57, 3.20)	<0.0001
Latino men	3.27 (3.06, 3.51)	<0.0001
White women	2.82 (2.73, 2.93)	<0.0001
White men	2.78 (2.70, 2.87)	<0.0001

1) MAT = Medication-assisted therapy; AI/AN = American Indian/Alaska Native; aOR = adjusted odds ratio;

2) Non-heroin includes prescription opioids, non-prescription methadone, or any other opioids with morphine-like effects

3) Model adjusted for age group, educational attainment, outpatient service type, treatment episode duration, opioid type, injection status, co-occurring alcohol abuse, and year

MAT was significantly and positively associated with a decrease in opioid use across all groups. The effect of MAT varied widely between racial/ethnic/gender groups, demonstrating within-group effects of moderation. The strongest effects were observed among AI/AN women and AI/AN men. While TEDS-D generally does not include data on facilities operated by federal agencies, some facilities that are operated by the IHS are included in the dataset. Our strong AI/AN finding could be partially explained by a recent and concerted effort by the IHS to invest substantial resources into improving OUD treatment outcomes (Duvivier et al. 2017). Furthermore, AI/ANs are more likely to forego substance use treatment when IHS services are unavailable (Fox 2011). In a sample of AI/ANs in the Midwest, Walls et al. (2006) also identified that IHS- or reservation-based treatment services were strongly preferred, as were treatment services that incorporated traditional and cultural elements into service delivery. Similarly, Redden, Tracy, and Shafer (2013) demonstrated a preference for a more cultural and spiritual framing of opioid addiction and MAT participation among AI/AN. While the development of evidence-based, culturally appropriate treatment strategies in AI/AN communities is ongoing (Novins et al. 2011), AI/AN-specific MAT services may be more effective in reducing opioid use than services available to other racial/ethnic groups. There may also be other social or environmental factors that affect the outpatient MAT experience differently between groups, including interpersonal provider/client communication or access to other contemporaneous health services. Therefore, our findings warrant

a discussion of the broader treatment experiences underlying disparities in the effect sizes of MAT.

Physicians' perceptions of opioid withdrawal severity – a key factor in prescribing MAT – differ based on the racial/ethnic or gender group of the client. For example, some clinics with a higher percentage of Black MAT patients are more likely to dispense methadone doses lower than the recommended amount, compared to predominantly White-attended clinics (D'Aunno and Pollack 2002). Importantly, lower methadone doses dispensed in some predominantly Black-attended clinics have been accompanied by factors that enable treatment success, such as higher levels of cultural competency among providers and addressing ancillary employment and housing services (Howard, Barrett, and Holmes 2010). Thus, low methadone doses alone may not be sufficient in predicting treatment outcomes. Furthermore, our findings that the positive effect sizes of MAT were particularly strong among racial/ethnic minority treatment participants complement the broad conclusion of Krawczyk et al. (2017) that MAT for OUD may be a unique clinical scenario where Whites fare worse than their nonwhite counterparts.

Finally, private office-based OUD treatment clients are more likely to be White (Lagisetty et al. 2019); however, non-subsidized private clinics do not report to TEDS. It is likely that private White MAT clients have better outcomes than their counterparts in state or federally subsidized treatment settings. In addition, other treatment clients may receive MAT services in non-substance abuse clinics and will not be recorded in TEDS, particularly if they receive MAT through primary care service providers (Watkins et al. 2017). Therefore, our findings of differential associations based on race/ethnicity/gender are only generalizable to treatment clients in non-private settings.

Limitations

Our definition of MAT included a single grouping of methadone or buprenorphine. However, these medications differ in several ways; our findings of race/ethnicity/gender disparities may be driven by one medication or the other. Future research may address the differences in methadone and buprenorphine by disaggregating findings by specific MAT regimen.

We did not include an indicator of opioid problem severity in our analyses, but problem severity likely confounds the relationship between MAT and a reduction in opioid use. TEDS-D includes a variable of substance abuse or dependence, based on definitions

in the Diagnostic and Statistical Manual of Mental Disorders-IV (DSM-IV), but the majority (92%) of treatment episodes are missing this data point. Rather than assessing only treatment episodes with non-missing DSM-IV diagnostic information, we opted to leave this variable out of analyses.

There is inconsistency as to which states report treatment data to SAMHSA each year. Georgia, Oregon, and West Virginia did not report data in 2015, 2016, or 2017, so were excluded from our analyses. Our final analytic sample included data from 47 states, Washington, D.C., and Puerto Rico.

Conclusion

MAT was positively associated with a reduction in opioid use between treatment admission and discharge. This relationship was moderated by race/ethnicity/gender, such that the positive effect sizes were strongest among AI/AN women and AI/AN men, and weakest among White men and White women. Growing evidence supports the perspective that the opioid epidemic and the resulting treatment demand may be disproportionately disaffecting White treatment clients. Conversely, strengthened IHS treatment services may be facilitating improved outcomes among AI/AN. Differential MAT outcomes between racial/ethnic and gender groups are clear, and are affected by multiple factors related to OUD treatment. OUD treatment services cater to a diverse population, and clinical MAT decisions should incorporate a comprehensive understanding of disparate racial/ethnic and gender outcomes.

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